

model-access-architecture

Model-Access Architecture — Diagrams (Key-Recognition · Unified Catalog · CLI-Bridge)

Field	Value
Revision	1
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Status	DESIGN — describes target architecture; nodes labelled IMPLEMENTED vs PLANNED
Maintainer	DOCUMENTATION subagent (READ-ONLY on code)
Authority	Cascades from constitution/ + root CLAUDE.md/CONSTITUTION.md; CONST-036 (verifier single source of truth), CONST-040 (capabilities from VerificationResult), §11.4 anti-bluff
Sources	SP1 plan (docs/superpowers/specs/plans/2026-06-10-SP1-model-access-plan.md), SP2 plan (.../2026-06-10-SP2-helixagent-exposure-plan.md), SP4 plan (.../2026-06-10-SP4-cli-bridge-plan.md); analyses B/C/D (docs/superpowers/specs/analysis/2026-06-10-{B,C,D}-*.md); master roadmap (docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md)

§11.4 / §11.4.123 anti-bluff honesty note (read first). This document describes the **target architecture** of the model-access + provider-

exposure + CLI-bridge system. It mixes **IMPLEMENTED** components (real code, cited `file:line`, confirmed in the analysis passes) with **PLANNED** components (specified in SP1/SP2/SP4 but **not yet built** — no captured runtime evidence exists). Every diagram node carries an explicit `[IMPLEMENTED ...]` or `[PLANNED ...]` tag. A node tagged **PLANNED** is a design intent, **not** a claim of working software. Diagrams are documentation, not proof of behaviour; the runtime-evidence bar (§11.4.5 / §11.4.69 / §11.4.107) is met only by the test/Challenge evidence each SP plan requires at execution time.

Legend used in every diagram:

- **IMPLEMENTED** — real code exists today; cited to a real `file:line` read during the 2026-06-10 analysis passes.
- **PLANNED** — specified in an SP plan / task (e.g. SP1 · T1.2.1); **not built**; no runtime evidence.
- **DEAD/UNWIRED** — code exists but has zero production call-sites (a §11.4.124 investigate-then-wire target, **not** removed).

Paths: inner Go app code is under `helix_code/` (module `dev.helix.code`); the HelixAgent exposure surface is under `submodules/helix_agent/` (module `dev.helix.agent`).

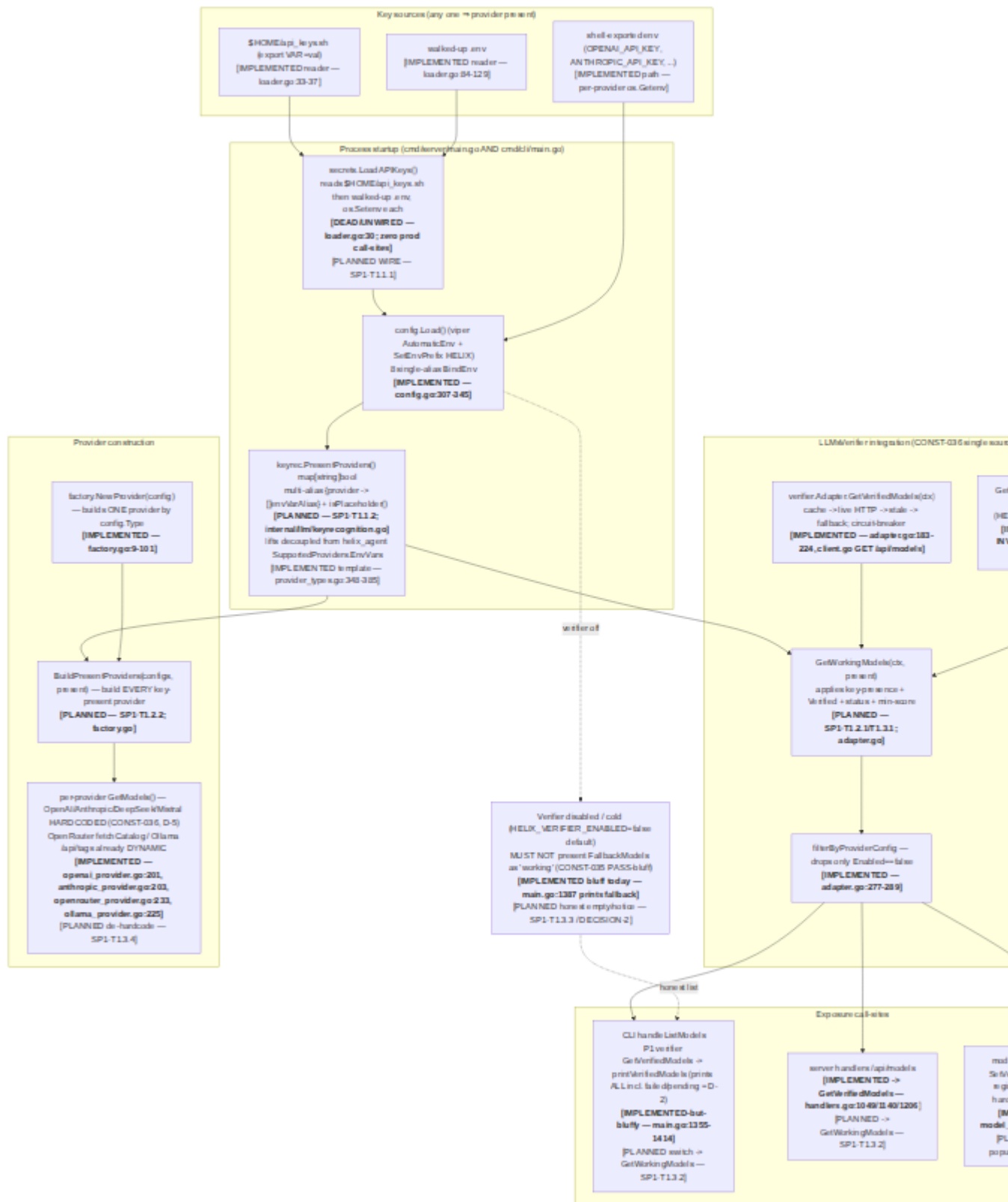
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Diagram A — Key-recognition → verifier → working-model funnel (SP1)

What it shows. How a single supplied/recognized API key turns into the set of WORKING models a user sees. Source: SP1 plan §3 “Target design (the working-model funnel)”; analysis-B §4.1.

Working-model predicate (SP1 §3, applied in the PLANNED `GetWorkingModels`): `present[provider] ∧ Verified==true ∧ VerificationStatus=="verified" ∧ OverallScore >= GetMinAcceptableScore() ∧ Providers[provider].Enabled != false.`



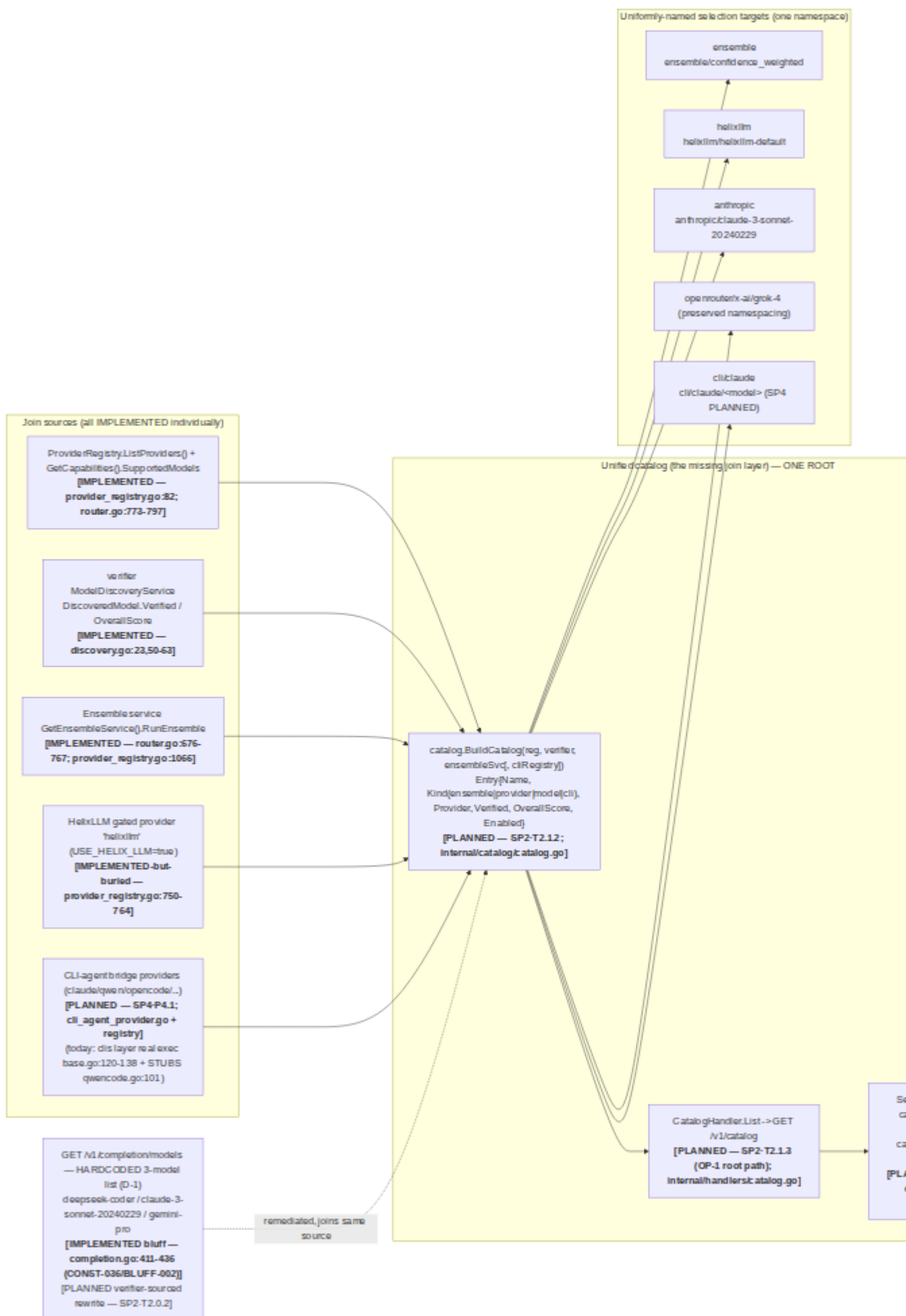
Mermaid diagram 1

Defect anchors closed by SP1 (analysis-B top-5 gaps): D-2 (CLI shows failed/pending), D-3 (LoadAPIKeys dead), D-4 (min-score never applied), D-5 (hardcoded lists), plus the no-key-presence-gate gap and the single-provider-factory gap.

Diagram B — Unified catalog under one root (SP2 + SP4)

What it shows. The PLANNED unified catalog that joins, **under one root**, the AI-debate ensemble + HelixLLM + every discovered provider + every provider's verified model + (SP4) every CLI-agent bridge provider and its models — each a uniformly-named, individually-addressable selection target. Source: SP2 plan §P2.1/§P2.2; analysis-C §3; master roadmap §257; SP4 plan §P4.1/§P4.3.

Naming grammar (analysis-C §3.2, master roadmap line 257): ensemble · ensemble/<preset> · helixllm · helixllm/<model> · <provider> · <provider>/<model_id> (preserving already-namespaced ids like openrouter/x-ai/grok-4) · **(SP4)** cli/<agent> · cli/<agent>/<model>.



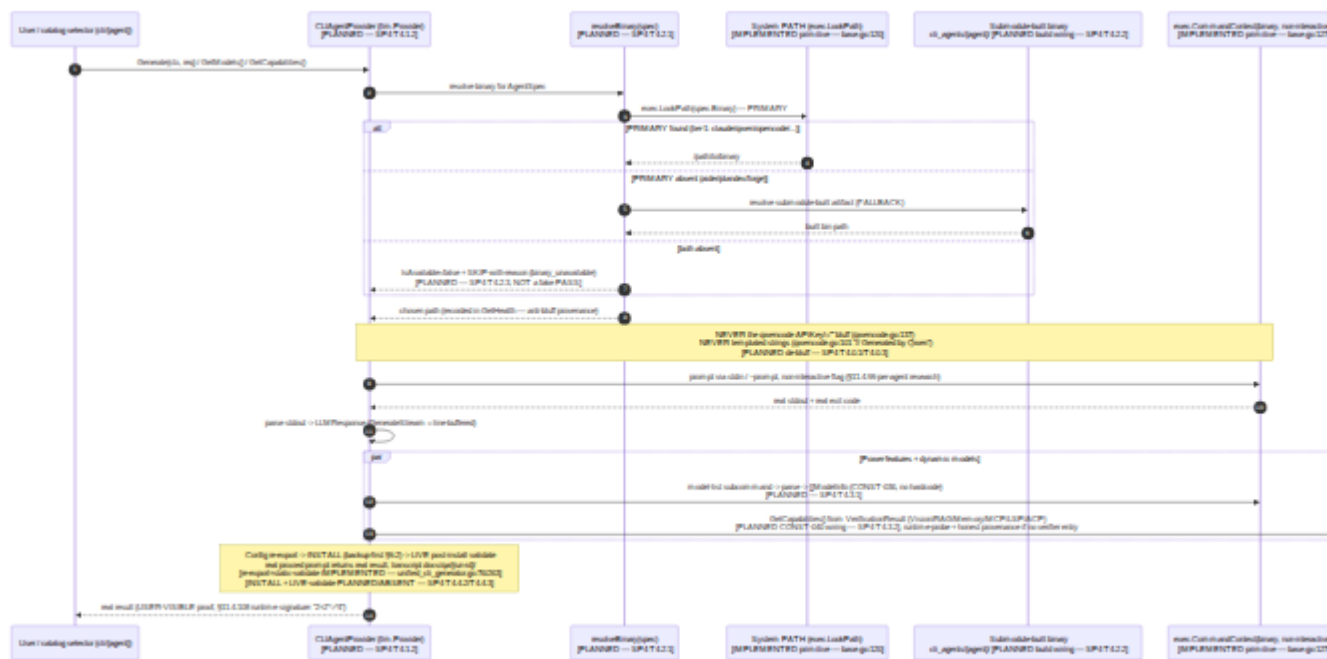
Mermaid diagram 2

Anti-bluff invariant (SP2 §7). Catalog “model” entries are emitted **only** for `DiscoveredModel.Verified == true` — never the static `SupportedModels` slice, never a hardcoded literal. HelixLLM is **promoted, never removed** (§11.4.122); all existing `/v1/*` routes stay intact (additive design).

Diagram C — CLI-bridge selection & proxy sequence (SP4)

What it shows. How a `cli/<agent>` selection resolves a binary (PRIMARY system-installed → FALLBACK submodule-built), proxies a prompt via real exec, parses the result, and surfaces power-features. Source: SP4 plan §P4.2/§P4.3/§P4.4; analysis-D §5.

Tier-1 agents system-installed today (analysis-D §2, captured command -v): claude 2.1.170, qwen 0.5.0, opencode 1.16.2, gemini 0.1.9, crush v0.22.2, codex, goose 1.8.0, copilot 1.0.46. **Fallback-required (absent today):** aider, plandex, forge.



Mermaid diagram 3

Wiring prerequisite (SP4 §2.3, confirmed gap D-7). `helix_code/go.mod` (replace block lines 194-212) has **no** `replace dev.helix.agent` — so `HelixCode` cannot import the `cli` substrate today. SP4-T4.0.5 PLANS to add it (or a narrow shim, decided on `go mod graph` evidence). The `exec` primitive itself (base.go:120-138) is **IMPLEMENTED** and real; the per-agent dispatch tier (`instance_manager.go:906+ executeClaudeCode/...`) and

qwencode.go:101/115/137 are **STUBS** (a §11.4 bluff SP4·P4.0 RED-reproduces then fixes).

Node provenance ledger (IMPLEMENTED vs PLANNED)

A single-table audit so a reader can verify each claim against real code or a real SP task.

Component	State	Anchor (real file:line) or
secrets.LoadAPIKeys reader	IMPLEMENTED but DEAD/ UNWIRED	helix_code/internal/secret_loader.go:30,33-37,84-1
Wire LoadAPIKeys at startup	PLANNED	SP1·T1.1.1 (cmd/server/main/cli/main.go)
Viper AutomaticEnv + 8 BindEnv	IMPLEMENTED	helix_code/internal/config/config.go:307-345
Multi-alias key-recognition table	PLANNED	SP1·T1.1.2 (internal/llm/keyrecognition.go)
helix_agent SupportedProviders.EnvVars (reuse template)	IMPLEMENTED	submodules/helix_agent/verifier/provider_types
verifier.Adapter.GetVerifiedModels (real HTTP)	IMPLEMENTED	helix_code/internal/verifier_adapter.go:183-224; cli/api/models
GetMinAcceptableScore (loaded, never invoked)	IMPLEMENTED- unused (D-4)	helix_code/internal/verifier_adapter.go:175-180
GetWorkingModels filter	PLANNED	SP1·T1.2.1 / T1.3.1 (adapter.
filterByProviderConfig (Enabled flag only)	IMPLEMENTED	helix_code/internal/verifier_adapter.go:277-289
CLI handleListModels / printVerifiedModels (shows failed/ pending, D-2)	IMPLEMENTED- bluffy	helix_code/cmd/cli/main.go:1355-1414
Switch CLI/server/registry → GetWorkingModels	PLANNED	SP1·T1.3.2
Verifier-disabled honest behaviour	PLANNED	SP1·T1.3.3 / DECISION-2 (today main.go:1387)

Component	State	Anchor (real file:line) or
factory.NewProvider (one by config.Type)	IMPLEMENTED	helix_code/internal/llm/factory.go:9-101
BuildPresentProviders enumerator	PLANNED	SP1·T1.2.2
Hardcoded provider model lists (D-5/CONST-036)	IMPLEMENTED (violation)	openai_provider.go:201, anthropic_provider.go:201, deepseek_provider.go, mistral_provider.go
OpenRouter fetchCatalog / Ollama / api/tags (dynamic templates)	IMPLEMENTED	openrouter_provider.go:225, ollama_provider.go:225
ProviderRegistry.ListProviders + GET /v1/providers	IMPLEMENTED	submodules/helix_agent/services/provider_registry/internal/router/router.go:676-76
Ensemble route POST /v1/ensemble/completions	IMPLEMENTED	submodules/helix_agent/router/router.go:676-76
HelixLLM gated provider helixllm (buried)	IMPLEMENTED	submodules/helix_agent/services/provider_registry.go:75
DiscoveredModel.Verified / OverallScore (working flag)	IMPLEMENTED	submodules/helix_agent/verifier/discovery.go:5
GET /v1/completion/models hardcoded 3-model list (D-1)	IMPLEMENTED bluff	submodules/helix_agent/handlers/completion.go:101, 115, 137
D-1 verifier-sourced rewrite	PLANNED	SP2·T2.0.2
catalog.BuildCatalog join	PLANNED	SP2·T2.1.2 (internal/catalog/catalog.go)
GET /v1/catalog handler + route	PLANNED	SP2·T2.1.3 (OP-1 root path)
Selector resolution (Name → target)	PLANNED	SP2·T2.1.4
HelixLLM first-class root promotion	PLANNED	SP2·T2.2.4
llm.Provider contract (CLI bridge target)	IMPLEMENTED	helix_code/internal/llm/missing_types.go:356-37
CLI-exec primitive (LookPath + CommandContext)	IMPLEMENTED	submodules/helix_agent/clis/agents/base/base.go:101, 115, 137
qwencode stub (templated strings, APIKey gate)	IMPLEMENTED bluff (D-6)	submodules/helix_agent/clis/agents/qwencode/qwencode.go:101, 115, 137
instance_manager dispatch stubs		

Component	State	Anchor (real file:line) or
	IMPLEMENTED bluff (wider than D-6)	submodules/helix_agent/ clis/instance_manager.g
replace dev.helix.agent wiring (D-7)	PLANNED/ ABSENT	SP4·T4.0.5 (helix_code/go. has no such replace)
CLIAgentProvider + AgentSpec registry	PLANNED	SP4·T4.1.2 / T4.1.3
resolveBinary PRIMARY → FALLBACK	PLANNED	SP4·T4.2.1
Dynamic GetModels per agent (CONST-036)	PLANNED	SP4·T4.3.1
Capabilities from VerificationResult (CONST-040)	PLANNED	SP4·T4.3.2
Config re-export + static Validate()	IMPLEMENTED	submodules/helix_agent/ unified_cli_generator.g cmd/helixagent/main.go:
Filesystem INSTALL + LIVE post-install validate	PLANNED/ ABSENT	SP4·T4.4.2 / T4.4.3

Cross-references

- **SQL data model** for these flows: [model-access-schema.sql](#).
- **Index / metadata header**: [README.md](#).
- **SP plans (authoritative task source)**: docs/superpowers/specs/plans/2026-06-10-SP1-model-access-plan.md, ...-SP2-helixagent-exposure-plan.md, ...-SP4-cli-bridge-plan.md.
- **Analysis (evidence base)**: docs/superpowers/specs/analysis/2026-06-10-B-model-access.md, ...-C-helixagent-exposure.md, ...-D-cli-bridge.md.
- **Master roadmap**: docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md.

Per CONST-066 / §11.4.65, this .md should ship synchronized .html/.pdf siblings at release-prep. They are **not** generated by this READ-ONLY documentation pass.